Stat 293: Tutorial 6

Q1 A plant scientist is interested in investigating the relationship between temperature and the percentage of fungal spores that germinate. She selects four temperatures (15, 20, 25 and 30◦ Celsius) and incubates four randomly selected microscopic slides at each of the four temperatures. She selects 10 locations on each slide at each of the four temperatures and determines the percentage of spores germinated.

(a) Identify the design, factors and number of levels of each factor in this experiment.

This is a nested design as each slide is incubated at only one temperature

Response Variable: % of Fungal Spores that Germinate
Factor A: Temperature (fixed effect)
Factor A Levels: {15^0C, 20^0C, 25^0C, 30^0C} -> a = 4
Factor B: Slide (nested inside Temperature) (random effect)
Factor B Levels: {Slide1, Slide2, Slide3, Slide4} -> b = 4
10 replicates -> n = 10

(b) Explain why microscopic slide should be fitted as a random effect.

Each slide is selected at random from all possible slides. The scientist has no particular interest in the specific slide used.

(c) Write the model equation and its constraints and assumptions.

$$\boldsymbol{Y_{ijk} = \mu + \alpha_i + B_{j(i)} + \epsilon_{k(ij)} \quad \forall\, i = 1, \dots, 4 \quad j = 1, \dots, 4 \quad k = 1, \dots, 10}$$

Where $\boldsymbol{\epsilon_{k(ij)} \sim N(0, \sigma^2)}$ and $\boldsymbol{B_{j(i)} \sim N(0, \sigma_B^2)}$ and we have the constraint $\boldsymbol{\sum_{i=1}^{4} \alpha_i = 0}$

Q2 Andrew & Underwood (1993) studied the effects of sea urchin grazing on a shallow subtidal reef in New South Wales, Australia. They set up four urchin density treatments (0% original, 33% original, 66% original, 100% original), with four patches of reef being randomly selected for each treatment and five quadrats selected from each patch. The response variable was % cover of filamentous algae in each quadrat. The data are available in the file andrew.csv.

(a) Identify the main treatment variable, A and the nested variable, B, in this experiment. Give the number of levels of each of these variables.

This is a nested design as each patch of reef is given only one treatment

Response Variable: % cover of filamentous algae
Factor A: Treatment (fixed effect)
Factor A Levels: {0% original, 33% original, 66% original, 100% original} -> a = 4
Factor B: Patch (nested inside Treatment) (random effect)
Factor B Levels: {Patch1, Patch2, Patch3, Patch4} -> b = 4
5 replicates -> n = 5

(b) Write down the model equation and its constraints and assumptions.

$$\boldsymbol{Y_{ijk} \quad = \quad \mu \; + \; \alpha_i \; + \; B_{j(i)} \; + \; \epsilon_{k(ij)} \qquad \forall\, i = 1, \dots, 4 \quad j = 1, \dots, 4 \quad k = 1, \; \dots, \; 5}$$

Where $\boldsymbol{\epsilon_{k(ij)} \sim N(0, \sigma^2)}$ and $\boldsymbol{B_{j(i)} \sim N\left(\, 0, \; \sigma_B^2 \,\right)}$ and we have the constraint $\boldsymbol{\sum_{i=1}^{4} \alpha_i = 0}$

(c) Explain why the nested variable should be fitted as a random effect.

The patches of reef represent a random sample of all possible patches. The researchers have no particular interest in the specific patch.

(d) The data were analysed using the lme() function in R. Use the output shown below to estimate the proportion of variation due to patch differences after adjusting for treatments.

```
seaurchin<-read.csv("andrew.csv",header=T)
str(seaurchin)

##'data.frame': 80 obs. of 4 variables:
## $ treat: Factor w/ 4 levels "con","rem","t0.33",..: 1 1 1 1 1 1 1 1 1 1 ...
## $ patch: int 1 1 1 1 1 2 2 2 2 2 ...
## $ quad : int 1 2 3 4 5 1 2 3 4 5 ...
## $ algae: int 0 0 0 6 2 0 0 0 0 0 ...

seaurchin$patch<-as.factor(seaurchin$patch)

library(nlme)
fit1<-lme(algae ~ treat, random=~1|patch, data=seaurchin)
summary(fit1)


## Linear mixed-effects model fit by REML
## Data: seaurchin
##        AIC        BIC        logLik
##        694.1502   708.1346   -341.0751
##

## Random effects:
## Formula: ~1 | patch
##              (Intercept)   Residual
## StdDev:      17.15554      17.28005

##

## Fixed effects: algae ~ treat
##              Value   Std.Error     DF   t-value      p-value
##(Intercept)   1.30    9.407876      64   0.1381821    0.8905
## treatrem 3   7.90    13.304746     12   2.8486075    0.0147
## treatt0.33   17.70   13.304746     12   1.3303523    0.2081
## treatt0.66   20.25   13.304746     12   1.5220132    0.1539
```

```
anova(fit1)
##              numDF  denDF   F-value        p-value
##(Intercept)   1      64      18.555081      0.0001
##treat         3      12      2.717102       0.0913
```

$\sigma_B^2 = 17.15554^2 \quad \sigma^2 = 17.28005^2$

$$=> \frac{\sigma_B^2}{\sigma_B^2 + \sigma^2} = \frac{17.15554^2}{17.15554^2 + 17.28005^2} = 49.6\%$$

(e) The data were also analysed using the aov() function. Use the output shown below to confirm that the variance components you get from aov match those from lme.

```
##Error: treat:patch
##              Df     Sum Sq  Mean Sq      F value  Pr(>F)
## treat        3      14429   4810         2.717    0.0913 .
## Residuals    12     21242   1770
##---
##Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
##Error: Within
##              Df     Sum Sq  Mean Sq      F value  Pr(>F)
##Residuals     64     19110   298.6
```

$MSE = \widehat{\sigma}^2 = 298.6 = 17.28005^2$ as before

$MS_{B(A)} = \widehat{\sigma}^2 + 5\,\hat{\sigma}_B^2 = 1770$

$=> \hat{\sigma}_B^2 = \frac{1770 - \widehat{\sigma}^2}{5} = 294.3 = 17.15554^2$ as before

$$=> \frac{\sigma_B^2}{\sigma_B^2 + \sigma^2} = \frac{17.15554^2}{17.15554^2 + 17.28005^2} = 49.6\%$$

(f) Is there evidence of differences among treatments? State the hypotheses, test statistic, p−value and conclusion.

$H_0 : \alpha_i = 0 \quad \forall\, i = 1, \ldots, 4$

$H_1 : \alpha_i \neq 0 \quad for\ some\ i$

F=2.717 with 3 and 12 degrees of freedom.
P-value = 0.0913
=> Weak evidence against the null hypothesis, i.e. weak evidence that the mean algae cover differs across urchin treatments.

Q3 A study was carried out to determine whether there is an effect of schools on standardized test scores and whether the effect of school is driven in part by differences in teachers. The data collected are shown below.

School 1		School 2		School 3	
Teacher 1	Teacher 2	Teacher 3	Teacher 4	Teacher 5	Teacher 6
25	14	11	22	17	5
29	11	6	18	20	2

(a) Write down the model equation for a nested design as it would apply in this case.

$$\boldsymbol{Y_{ijk} = \mu + \alpha_i + B_{j(i)} + \epsilon_{k(ij)} \quad \forall\, i = 1, \dots, 3 \quad j = 1, 2 \quad k = 1, 2}$$

Where $\boldsymbol{\epsilon_{k(ij)} \sim N(0, \sigma^2)}$ and $\boldsymbol{B_{j(i)} \sim N(0,\ \sigma_B^2)}$ and we have the constraint $\boldsymbol{\sum_{i=1}^{3} \alpha_i = 0}$

(b) Using the model in part (a) above, work out $E[Y_{ijk}]$ and $Var[Y_{ijk}]$.

$$E[\boldsymbol{Y_{ijk}}] = \boldsymbol{E[\mu + \alpha_i + B_{j(i)} + \epsilon_{k(ij)}]}$$

$$= \boldsymbol{E[\mu] + E[\alpha_i] + E[B_{j(i)}] + E[\epsilon_{k(ij)}]}$$
$$= \mu + \alpha_i$$

$$Var[\boldsymbol{Y_{ijk}}] = \boldsymbol{Var[\mu + \alpha_i + B_{j(i)} + \epsilon_{k(ij)}]}$$

$$= \boldsymbol{Var[\mu] + Var[\alpha_i] + Var[B_{j(i)}] + Var[\epsilon_{k(ij)}]}$$
$$= \sigma_B^2 + \sigma^2$$

(c) Enter the data into Excel, ensuring that you use unique labels for any nested factors there may be. Import the data into R and use the aov function to carry out an ANOVA test for mean score differences between schools.

```
# 3c.1 - Read in the data
sch1a <- read.csv("F:\\Lecture Courses\\STAT 293\\...\\ school.csv")
str(sch1a)
```

```
## 'data.frame': 12 obs. of  3 variables:
## $ School : chr  "S1" "S1" "S1" "S1" ...
## $ Teacher: chr  "T1" "T1" "T2" "T2" ...
## $ Score  : int  25 29 14 11 11 6 22 18 17 20 ...
```

```
# 3c.2 - Fitting the model
sch1b <- aov(Score ~ School + Error(School:Teacher) , data=sch1a)

# 3c.3 - Obtaining Summary Statistics
summary(sch1b)
```

Error: School:Teacher

##	Df	Sum Sq	Mean Sq	F value	Pr(>F)
## School	2	156.5	78.25	0.414	0.694
## Residuals	3	567.5	189.17		

Error: Within

##	Df	Sum Sq	Mean Sq	F value	Pr(>F)
## Residuals	6	42	7		

$H_0 : \alpha_i = 0 \quad \forall\, i = 1, \ldots, 4$

$H_1 : \alpha_i \neq 0 \quad for\ some\ i$

F=0.414 with 2 and 3 degrees of freedom.
P-value = 0.694
=> No evidence against the null hypothesis, i.e. There is no evidence that mean scores differ across schools

(d) Use output from the aov function to estimate the variance components σ_B^2 and σ^2.

$MSE = \widehat{\sigma}^2 = 7$

$MS_{B(A)} = \widehat{\sigma}^2 + 2\,\hat{\sigma}_B^2 = 189.17$

$$=> \hat{\sigma}_B^2 = \frac{189.17 - \widehat{\sigma}^2}{2} = 91.085$$

(e) Within schools, what proportion of variation in scores is due to teachers and what proportion is due to students?

$=> \frac{\sigma_B^2}{\sigma_B^2 + \sigma^2} = \frac{91.085}{91.085 + 7} = 92.9\%$ proportion of variation due to teachers

$\frac{\sigma^2}{\sigma_B^2 + \sigma^2} = \frac{7}{91.085 + 7} = 7.1\%$ proportion of variation unexplained (i.e. due to students)